A Comparative Analysis of Resampling Techniques for Addressing Class Imbalance in Multiclass Classification

1st Muhammad Asif
Faculty of Computing and Applied Sciences
Sir Syed University of Engg. & Tech.,
Karachi, Pakistan
muasif@ssuet.edu.pk

2nd Jawad Ali Bhatti
Dept. of Electronic Engineering
Sir Syed University of Engg. & Tech.,
Karachi, Pakistan
jabhatti@ssuet.edu.pk

3rd Muhammad Safwan
Dept. of Electronic Engineering
Sir Syed University of Engg. & Tech.,
Karachi, Pakistan
msafwan@ssuet.edu.pk

4th Muhammad Rafay Khan
Dept. of Electronic Engineering
Sir Syed University of Engg. & Tech.,
Karachi, Pakistan
murafay@ssuet.edu.pk

5th Fahad Ahmed Siddiqui
Dept. of AC&ES
Sir Syed University of Engg. & Tech.,
Karachi, Pakistan
fahmeds@ssuet.edu.pk

6th Muhammad Rehan
Dept. of Electronic Engineering
Sir Syed University of Engg. & Tech.,
Karachi, Pakistan
murehan@ssuet.edu.pk

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***Abstract*—This study addresses the issue of class imbalance in multiclass time series classification by using various resampling techniques. In this study, four prominent strategies are used that are Synthetic Minority Over-sampling Technique (SMOTE), Adaptive Synthetic (ADASYN) sampling, Random Under-sampling, and a hybrid method (SMOTETomek). The techniques are applied to the MIT-BIH Arrhythmia Dataset as a case study. The performance and impact of each technique are measured on classification performance. Our methodology employs a Random Forest classifier and a comprehensive set of evaluation metrics. Results demonstrate that while all resampling methods improve minority class recognition, the hybrid SMOTE-Tomek approach shows the most promising results, achieving the best balance between overall performance and class-wise metrics.**

***Index Terms*—Class imbalance, Multiclass classification, Resampling techniques, SMOTE, ADASYN, Random Undersampling, SMOTETomek, ECG classification**

I. Introduction

Class imbalance in datasets is one of the significant problem to solve as its affects the performance of AI and machine learning algorithms, especially in multiclass classification [1]. This imbalance, which is characterized by a skewed distribution of instances across classes, leads to some classes being significantly underrepresented in the dataset. The implications of class imbalance are extensive, affecting model training, performance evaluation, and the practical application of classification systems in real-world scenarios [2]. In applications such as medical diagnosis, fraud detection, and prediction of rare events, class imbalances are often an inherent feature of data [3]. The challenges posed by class imbalance are multifaceted [4], [6]:

- Machine learning and AI algorithms show a bias towards the class that are in a majority, often leading to misclassifying or overlooking minority class samples [5]. This bias arises from the optimization objective of these algorithms, which is usually aimed at maximize overall accuracy.
- Traditional performance metrics such as accuracy can be misleading when applied to imbalanced datasets [1]. A model achieving 99% accuracy may appear impressive, but if the minority class represents only 1% of the data, this model can predict the majority class for all instances, completely failing to identify the critical minority class.

A. Literature Review

In recent years, there has been significant progress in addressing class imbalance. Eltayeb et al. [12] conducted a comprehensive review of resampling techniques, categorizing them into three main groups: oversampling, undersampling and hybrid methods that is the combination of both oversampling and undersampling [6].

Oversampling technique such as Synthetic Minority Oversampling Technique (SMOTE), introduced by [5], is perhaps the most widely adopted and various variants have also been introduced over the years.

SMOTE generates synthetic samples for the classes that are in minority by interpolating between existing instances. He et al. [8] proposed the Adaptive Synthetic (ADASYN) sampling approach, which extends SMOTE by focusing on generating samples near the decision boundary.

Undersampling techniques balance the dataset by reducing the number of instances of the classes that are in the majority [7]. Although effective, these methods run the risk of missing essential information from the majority class. Interestingly, Tarawneh et al. [14] argue against oversampling, highlighting potential issues such as overfitting and increased computational complexity.

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Hybrid methods, combining both undersampling and oversampling, have shown promise in recent studies. Batista et al. [9] introduced SMOTETomek, which applies SMOTE followed by Tomek Links removal, aiming to create well-defined class clusters.

In the context of time series data, which is particularly relevant to our study, Moniz et al. [15] and Zhu et al. [16] have proposed specialized resampling strategies for imbalanced time series classification and forecasting problems.

López et al. [4] conducted an extensive study on the effectiveness of various resampling techniques across different domains. Their findings, echoed by Haixiang et al. [13], suggest that the performance of resampling methods can vary significantly depending on the dataset characteristics and the chosen classifier.

In the context of multiclass imbalance, Krawczyk [2] proposed a novel approach that decomposes the problem into multiple binary classification tasks, applying specific resampling techniques to each sub-problem. This strategy showed improved performance over traditional multiclass resampling methods in several benchmark datasets.

B. Our Approach

The study is focused on evaluating the effectiveness and performance of four resampling methods for addressing the class imbalance for multiclass classification problems. The four prominent strategies selected are SMOTE, ADASYN, Random Under-sampling, and SMOTETomek. The methodology applies various resampling techniques to address the class imbalance in a multiclass dataset for improving the classifier performance. We employ a Random Forest classifier, chosen for its robustness and its ability to manage multiclass datasets [10]. To evaluate the effectiveness of each resampling approach, we utilize multiple metrics: precision, recall, F1-score, and the area under the precision-recall curve (AUC-PR). To demonstrate the practical applicability, MIT-BIH Arrhythmia Database is used as a case study [11]. This dataset serves as an excellent example of real-world class imbalance in health sciences domain.

The rest of the paper is organized as follows: Section II gives a description of our methodology. Section III provides our results and discussion. Finally, Section IV concludes the paper, summarizing our findings and suggesting directions for future research.

Through the proposed study, we aim to contribute to the ongoing research in class imbalance mitigation, providing insights that can guide practitioners in selecting appropriate resampling techniques for imbalanced datasets across various domains.

II. Methodology

A. Problem Formulation

Class imbalance is a prevalent issue in multiclass time series classification, where the number of instances across different classes varies significantly. Let $\Omega = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$ represent our dataset, where $\mathbf{x}_i \in \mathbb{R}^{M \times L}$ denotes the i-th sample with M features across L time steps, and $y_i \in \{0, 1, \ldots, K-1\}$ is its corresponding class label. The imbalance ratio (IR) is defined as:

$$\text{IR} = \frac{\max_k N_k}{\min_k N_k} \tag{1}$$

where N_k is the number of instances in class k. Our objective is to mitigate this imbalance and improve classification performance, particularly for minority classes.

B. Data Preprocessing

a) Data Acquisition: For the proof of concept, we use the MIT-BIH Arrhythmia Database. The dataset consists of electrocardiogram (ECG) recordings, which we preprocess to create feature vectors for classification.

b) Loading and Splitting Data: The dataset is loaded and separated into training and test sets using stratified sampling to maintain the class distribution:

```
df = pd.read_csv('mitbih_train.csv',header=None)
X_train, X_test, y_train, y_test =
    train_test_split(X, y, test_size=0.2,
    random_state=42, stratify=y)
```

c) Feature Scaling: The features are scaled to ensure have a zero mean and a standard deviation of one, which is essential for the performance of many AI algorithms:

```
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

C. Class Imbalance Mitigation Techniques

We employ several resampling techniques to address the class imbalance issue. These techniques include oversampling, undersampling, and hybrid methods.

a) Oversampling: **Synthetic Minority Over-sampling Technique (SMOTE)**: This technique produces synthetic samples for the minority class by creating interpolations between existing samples within that class.

Adaptive Synthetic Sampling (ADASYN): ADASYN improves on SMOTE by generating synthetic samples that focus on difficult-to-learn areas, increasing the density of minority class samples near the decision boundary.

b) Undersampling: **Random UnderSampling**: This technique reduces the number of majority class samples by randomly selecting a subset, thereby balancing the class distribution.

c) Hybrid Methods: **SMOTETomek**: This method combines SMOTE with Tomek links to both over-sample the minority class and remove borderline majority class samples, enhancing class separation.

```
strategies = {
    'Original': None,
    'SMOTE': SMOTE(random_state=42),
    'ADASYN': ADASYN(random_state=42),
    'Undersampling':
        RandomUnderSampler(random_state=42),
    'Hybrid (SMOTETomek)':
        SMOTETomek(random_state=42)
}
```

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D. Model Training and Evaluation

a) Training the Model: For each resampling strategy, Random Forest classifier is trained on the resampled data.

```
for name, strategy in strategies.items():
    if strategy is not None:
        X_resampled, y_resampled =
            strategy.fit_resample
            (X_train_scaled, y_train)
    else:
        X_resampled, y_resampled =
            X_train_scaled, y_train

    clf = RandomForestClass(random_state=42)
    clf.fit(X_resampled, y_resampled)
```

b) Evaluation Metrics: We evaluate the model's performance using several metrics:

- Precision, Recall, and F1-Score: These metrics are calculated for each class and provide a balanced view of the classifier's performance.
- AUC-PR: The area under the precision-recall curve is used to evaluate the model's performance, especially for imbalanced datasets.
- Confusion Matrix: This gives a breakdown of true positives, false positives, true negatives, and false negatives for each class.

```
def evaluate_and_print(y_true,
        y_pred, y_prob, strategy_name):

    precision, recall, f1, _ =
        precision_recall_fscore_support
       (y_true, y_pred,average='weighted')

    cm = confusion_matrix(y_true, y_pred)
    auc_pr = average_precision_score(
        y_true, y_prob, average='macro')
```

c) Experimental Procedure: For each resampling strategy:

1) Apply the resampling strategy to the training data.
2) Training of a Random Forest classifier on the modified or resampled data.
3) Evaluate the classifier performance.
4) Record and analyze the performance metrics.

```
for name, strategy in strategies.items():
    if strategy is not None:
        X_resampled, y_resampled =
        strategy.fit_resample(X_train_scaled,
        y_train)

    else:
        X_resampled, y_resampled =
        X_train_scaled, y_train

    clf = RandomForestClass(random_state=42)
    clf.fit(X_resampled, y_resampled)

    y_pred = clf.predict(X_test_scaled)
    y_prob = clf.predict_proba(X_test_scaled)
    evaluate_and_print(y_test, y_pred,
    y_prob, name)
```

E. Case Study: MIT-BIH Arrhythmia Database

To demonstrate the effectiveness of our methodology, we apply it to the MIT-BIH Arrhythmia Database. The dataset contains ECG recordings classified into five classes: Normal beat (NB), Supraventricular premature beat (SPB), Premature ventricular contraction beat (PVC), Fusion of ventricular and normal beat (VNB), and Unclassifiable beat (UC). The class distribution in a dataset is highly imbalanced, making it an ideal candidate for applying our proposed resampling techniques.

This methodology provides a comprehensive framework for addressing class imbalance in multiclass time series classification. By employing a combination of oversampling, undersampling, and hybrid techniques, and evaluating their effectiveness using a robust set of metrics, we can enhance the classification performance, particularly for minority classes. The MIT-BIH Arrhythmia Database serves as a proof of concept to demonstrate the applicability of these techniques to real-world datasets.

III. Results and Discussion

This section shows the effectiveness of various resampling techniques in addressing class imbalance for the MIT-BIH Arrhythmia Database. We evaluated five approaches: the original imbalanced dataset, SMOTE, ADASYN, Undersampling, and a hybrid method (SMOTETomek).

A. Class Distributions

Table I shows the class distributions for the original dataset and after applying each resampling technique.

The original dataset exhibits severe class imbalance, with Normal beats vastly outnumbering other classes. SMOTE and SMOTETomek achieve near-perfect balance across all classes. ADASYN results in a slight variation in class sizes but still significantly improves balance. Undersampling achieves perfect balance but at the cost of substantial data reduction.

B. Performance Comparison

To evaluate the impact of these resampling techniques, accuracy, precision, recall, F1-score, and AUC-PR are used as metrics. The results are presented in Table II.

These results demonstrate that SMOTE, ADASYN, and SMOTETomek all improve the outcome of the classifier by addressing the class imbalance in the dataset. The hybrid SMOTETomek approach emerged as the most effective, achieving the best balance between overall performance and class-wise metrics.

C. Confusion Matrix Analysis

Examining the confusion matrices given in table III provides deeper insights into the performance of each method: **Original Dataset**: The model shows high accuracy for the majority class (Normal beat) with 14,467 correct predictions out of 14,494 instances. However, it struggles with minority classes, particularly the Supraventricular premature beat class, where it misclassifies 164 instances as Normal beats. **SMOTE**: This

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TABLE I
CLASS DISTRIBUTIONS BEFORE AND AFTER RESAMPLING

Class	Original	SMOTE	ADASYN	Undersampling	SMOTETomek
0: Normal beat (NB)	72,471	57,977	57,977	513	57,976
1: Supraventricular premature beat (SPB)	2,223	57,977	58,003	513	57,977
2: Premature ventricular beat (PVB)	5,788	57,977	57,874	513	57,976
3: Fusion of ventricular & normal beat (VNB)	641	57,977	57,998	513	57,977
4: Unclassified (UC)	6,431	57,977	57,984	513	57,977

TABLE II
PERFORMANCE METRICS BEFORE AND AFTER RESAMPLING

Metric	Original	SMOTE	ADASYN	Undersampling	SMOTETomek
Accuracy	97.52%	98.24%	98.18%	87.17%	98.31%
Precision	97.50%	98.19%	98.14%	94.19%	98.27%
Recall	97.52%	98.24%	98.18%	87.17%	98.31%
F1-Score	97.36%	98.20%	98.15%	89.78%	98.27%
AUC-PR	93.14%	94.13%	93.93%	86.51%	94.05%

method improves the recognition of minority classes. For instance, correct classifications of Supraventricular premature beats increased from 275 to 336, and Fusion beats from 76 to 94. There's a slight decrease in Normal beat accuracy (from 14,467 to 14,412), but this trade-off results in better overall performance. **ADASYN**: Similar to SMOTE, ADASYN improves minority class recognition. It performs particularly well on Supraventricular premature beats (357 correct classifications) and Unclassified beats (1,263 correct classifications), slightly outperforming SMOTE in these categories. **Undersampling**: While this method significantly improves the recognition of minority classes (e.g., 362 correct Supraventricular premature beats, 117 correct Fusion beats), it comes at a substantial cost to the majority class accuracy. Only 12,501 Normal beats are correctly classified, compared to 14,467 in the original dataset. **SMOTETomek**: This hybrid approach shows the best overall performance. It maintains high accuracy for Normal beats (14,417 correct) while significantly improving minority class recognition. For example, it correctly classifies 355 Supraventricular premature beats and 1,260 Unclassified beats.

Comparing the confusion matrices across all methods reveals that SMOTE, ADASYN, and SMOTETomek consistently reduced misclassifications for minority classes compared to the original dataset. The Undersampling method improved minority class recognition but at the cost of majority class performance.

In conclusion, this study highlights the importance of addressing class imbalance in ECG classification. The class distributions and performance metrics demonstrate the strengths and limitations of different resampling strategies. While all methods improved minority class recognition to some degree, the hybrid SMOTETomek approach showed the most promising results for balancing performance across all classes without sacrificing overall accuracy.

IV. CONCLUSION AND FUTURE WORK

This study has comprehensively examined the impact of various resampling techniques on addressing class imbalance in multiclass time series classification, using ECG data as a practical case study. Our findings underscore the critical importance of tackling class imbalance to enhance overall classification performance, particularly for minority classes that often represent critical but rare events in medical diagnostics.

The comparative analysis of SMOTE, ADASYN, Random Undersampling, and SMOTETomek revealed nuanced trade-offs between improving minority class recognition and maintaining majority class performance. While all resampling methods showed improvements over the original imbalanced dataset, the hybrid SMOTETomek approach emerged as the most promising. It achieved the best balance in performance across all classes, demonstrating that combining oversampling and undersampling techniques can yield superior results in complex, multiclass scenarios.

Our results highlight that selecting a resampling technique should be based on the specific features or characteristics of the dataset and the relative importance of different classes in the application domain. For instance, in ECG classification, where missing a rare arrhythmia could have severe consequences, techniques that boost minority class recognition without significantly compromising majority class accuracy, like SMOTETomek, are particularly valuable.

However, this study also reveals areas that warrant further investigation. Future work could explore several promising directions:

- Adaptive Resampling Strategies: Developing techniques that dynamically adjust the degree of oversampling or undersampling based on the characteristics of individual classes within a multiclass problem.
- Time Series-Specific Resampling: Creating or adapting resampling methods that specifically account for the

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TABLE III
CONFUSION MATRICES FOR DIFFERENT RESAMPLING TECHNIQUES

Confusion Matrix	NB	SPB	PVC	VNB	UC
Original Dataset					
NB	14467	8	13	2	4
SPB	164	275	5	0	1
PVC	120	1	1029	5	3
VNB	40	0	12	76	0
UC	55	0	2	0	1229
SMOTE					
NB	14412	38	26	12	6
SPB	82	336	6	0	1
PVC	64	1	1083	7	3
VNB	24	0	10	94	0
UC	26	0	3	0	1257
ADASYN					
NB	14383	44	45	8	14
SPB	83	357	5	0	0
PVC	52	1	1096	5	4
VNB	20	0	15	93	0
UC	19	0	4	0	1263
Undersampling					
NB	12501	665	416	823	89
SPB	68	362	8	5	2
PVC	44	14	1046	41	13
VNB	3	1	7	117	0
UC	34	3	10	1	1238
SMOTETomek					
NB	14417	37	26	8	6
SPB	85	355	5	0	0
PVC	58	2	1090	6	2
VNB	24	0	11	93	0
UC	24	0	2	0	1260

temporal nature of time series data, preserving important sequence information during the resampling process.

- Ensemble Methods: Investigating the potential of ensemble classifiers that combine multiple resampling techniques or integrate resampling with algorithm-level approaches to handle imbalance.
- Deep Learning Integration: Exploring how resampling techniques can be effectively integrated with deep learning models, particularly for complex time series data like ECGs.
- Cost-Sensitive Learning: Incorporating domain-specific misclassification costs into the resampling and classification process, especially crucial in medical applications where false negatives and false positives may have different implications.
- Scalability Analysis: Evaluating the computational efficiency and scalability of different resampling techniques on larger, more complex datasets to assess their practicality in real-world applications.
- Transfer Learning: Investigating how models trained on resampled data perform when applied to new, imbalanced datasets from similar domains.
- Interpretability: Developing methods to enhance the interpretability of models trained on resampled data, which is particularly important in healthcare applications where understanding model decisions is crucial.

By addressing these areas, future research can further refine and expand the toolkit available for handling class imbalance in multiclass time series classification. This ongoing work is essential for improving the reliability and effectiveness of machine learning models in critical domains such as healthcare, where accurate classification of rare events can have significant real-world impact.

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